[INCH-POUND]
A-A-59432A
9 December 2013
SUPERSEDING
A-A-59432
21 May 1999

COMMERCIAL ITEM DESCRIPTION

PLUGS, PIPE, MAGNETIC, HEADLESS: IRON, STEEL OR ALUMINUM ALLOY

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the general requirements for plugs, pipe, magnetic, headless: iron, steel or aluminum alloy. Plugs, pipe, magnetic, headless: iron, steel or aluminum alloy covered by this CID are intended for commercial/industrial applications.

2. CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This CID uses a classification system which is included in the PIN as shown in the following example (see 7.1).

3. CLASSIFICATION. Magnetic plugs shall be of the following types:

a. Dash 001: Type I - Aluminum alloy.
b. Dash 002: Type II - Malleable iron, cast iron, or cold finished steel.



3. SALIENT CHARACTERISTICS.

3.1 Interface and physical dimensions. Plugs, pipe, magnetic, headless: iron, steel or aluminum alloy supplied to this CID shall be as specified herein (see figure 1).

3.1.1 .Intended. These plugs, pipe, magnetic, headless: iron, steel or aluminum alloy are intended to cover: (1) headless magnetic pipe plugs having recessed hex heads, up to one inch size, as specified in SAE-J531, and (2) headless magnetic pipe plugs having recessed square heads, for sizes over one inch, as specified in ASME B16.14.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: DLA Land and Maritime, ATTN: VA**I**, P.O. Box 3990, Columbus OH 43218–3990, or email fluidflow@.dla.mil. Since contact information can change you may want to verify the currency of the address information using the ASSIST Online database at https://assist.dla.mil.

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3.1.2 Application. These magnetic pipe plugs are intended for use in automotive engine oil systems, transmissions, gear boxes, and circulating oil systems, and other systems of circulating fluids in which ferrous metal particles and/or chips may be present.

3.2 Design. The plugs, pipe, magnetic, headless: iron, steel or aluminum alloy design requirements for dimensions and tolerances, pipe thread size, and magnetic lift shall be in accordance with figure 1 and table I. The aluminum alloy plugs shall be anodized, in accordance with the contract or order. The ferrous plugs shall have a plain, uncoated finish, or other finish in accordance with the contract or order.

3.2.1 Material. The material for aluminum alloy plugs shall be in accordance table I. The material for malleable iron plugs shall be in accordance with ASTM A47/A47M. The material for cast iron plugs shall be in accordance with ASTM A48/A48M. The material for cold finish steel plugs shall be in accordance with ASTM A48/A48M

TABLE I. Aluminum alloys.

Temper	Specification
0, H12, H14, H16, H18, H112, or F	ASTM B211 alloy 1100
T3, T4, T8, or T451	ASTM B211 alloy 2011
T4	SAE-AMS-QQ-A-367 alloy 6061
0	SAE-AMS4115, alloy 6061
T4,T42	SAE-AMS4116 alloy 6061
T6, T62, T651	SAE-AMS4117, alloy, 6061
F	SAE-AMS4115, SAE-AMS4116, or SAE-AMS4117, alloy 6061
T451	SAE-AMS4128, alloy 6061

3.2.2 Finish. Anodize aluminum as specified in SAE-AMS2472 or MIL-A-8625, type I, class 1 or type II, class I. Ferrous plugs shall be plain (uncoated) finish.

3.2.3 Magnet capability. Magnetism shall be a permanent magnet in accordance with Magnetic Materials Producers Association (MMPA) Standard 0100. The magnet lift rating is defined as the weight of soft steel having one smooth side, that the magnet will retain after contact has been established, and for at least 48 hours after magnetization. The magnet shall be attached to the plug in such a manner as to require physical displacement of metal before the magnet can be removed. The maximum diameter of the magnet shall be at least 0.100 inch smaller than the pitch diameter of the plug thread.

3.3 Threads. The threads shall be NPTF for plugs up to and including the three (3) inch size and NPT for the four (4) inch size.

3.4 Marking. Plugs, pipe, magnetic, headless: iron, steel or aluminum alloy supplied to this CID shall be marked with the manufacturer's (MFR's) standard commercial PIN. (NOTE: The part number marked on the unit pack shall be the CID PIN.)

3.5 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.6 Workmanship. Plugs, pipe, magnetic, headless: iron, steel or aluminum alloy shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Certification. Certification must be done with the procuring activity approval. The contractor shall certify that the product offered meets the salient characteristics of the description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same as the product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See section 2 for PIN format example.

7.2 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmental Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website http://www.epa.gov/osw/hazard/wastemin/priority.htm. Included in the EPA list of 31 priority chemicals are cadmium, lead, and mercury. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein (see section 3).

7.3 Commercial and Government Entity (CAGE) code. For ordering purposes, inventory control, and submission of these plugs, pipe, magnetic, headless: iron, steel or aluminum alloy to DLA Land and Maritime under the Parts Management Advisory Team (PMAT), CAGE code 58536 should be used

7.4 Source of documents.

FEDERAL REGULATIONS

FAR – Federal Acquisition Regulations (FAR)

(Copies of these documents are available online at www.acquisition.gov/comp/far/index.html or from the U.S. Government Printing Office, 732 North Capital Street, NW, Washington D.C. 20401.)

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-A-8625 –

(Copies of these documents are available online at http://quicksearch.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

ASME INTERNATIONAL

ASME B16.14 – Ferrous Pipe Plugs, Bushings, and Locknuts with Pipe Threads

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(Copies of these documents are available online at http://www.asme.org or from the ASME International, Three Park Avenue, New York, NY 10016-5990.)

ASTM INTERNATIONAL

ASTM A47/A47M	–	Standard Specification for Ferritic Malleable Iron Castings
ASTM A48/A48M	–	Standard Specification for Gray Iron Castings
ASTM B211	–	Standard Specification for Aluminum and Aluminum-Alloy Rolled or Cold Finished Bar, Rod, and Wire

(Copies of these documents are available online at http://www.astm.org or from the ASTM International, P.O. Box C700, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

SAE-AMS2472	-	Anodic Treatment of Aluminum Alloys Sulfuric Acid Process, Dyed Coating
SAE-AMS4115	–	Aluminum Alloy, Rolled or Cold-Finished, Bars, Rods, Wire, and Flash Welded Rings Annealed 1.0Mg - 0.60Si - 0.28Cu - 0.20Cr (6061-0) - UNS A96061
SAE-AMS4116	–	Aluminum Alloy, Bars, Rods, and Wire 1.0Mg - 0.60Si - 0.30Cu - 0.20Cr (6061-T4) Cold Finished, Solution Heat Treated and Naturally Aged - UNS A96061
SAE-AMS4117	–	Aluminum Alloy, Rolled or Cold Finished Bars, Rods, and Wire and Flash Welded Rings 1.0Mg - 0.60Si - 0.28Cu - 0.20Cr (6061; -T6, -T651) Solution and Precipitation Heat Treated - UNS A96061
SAE-AMS4128	–	Aluminum Alloy Bars, Rolled or Cold Finished 1.0Mg - 0.60Si - 0.30Cu - 0.20Cr (6061-T451) Solution Heat Treated and Stress Relieved by Stretching - UNS A96061
SAE-AMS-QQ-A-367	–	Aluminum Alloy Forgings
SAE-J531	-	Automotive Pipe, Filler, and Drain Plugs

(Copies of these documents are available on line at www.sae.org from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, and Tel: 877-606-7323 [inside USA and Canada] or 724-776-4970 [outside USA], email at CustomerService@sae.org.)

Magnetic Materials Producers Association

MMPA STANDARD	-	100

Magnetic Materials Producers Association (MMPA), 8 SOUTH MICHIGAN ANENUE • SUITE 1000 • CHICAGO, ILLINOIS 60603

7.5 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.

b. Product conformance provisions.

c. Packaging requirements.

7.6 Commercial products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination,

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these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict acquisition to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
1MQB3	S I T CORPORATION 13200 SW 128TH ST BLDG H MIAMI, FL 33186-5881 Tel: (305) 255-3127 sitsyrny@twcny.rr.com Tel: (315) 422-5444 Otto@sitcorporation.com Tel: (305) 255-3127
3DPE3	JSO/MD,INC 1961 SAINT ANDREWS DR RED OAK, TX 75154-5837 Tel: (972) 617-0677 JSOMDinc@msn.com Tel: (469) 556-2805

MFR's CAGE	MFR's name and address – continued.
050R4	PATRIOT VALVE & FITTING CORP. 86 BROADWAY FREEPORT, NY 11520-2821 sales@patriotlogistics.com Tel: (516) 377-3300

7.7 Government users. To acquire information on obtaining these Plugs, pipe, magnetic, headless: iron, steel or aluminum alloy from the Government inventory system, contact DLA Land and Maritime, ATTN: DLA Land and Maritime Call Center (–NAB), P.O. Box 3990, Columbus, OH 43218–3990 or telephone (614) 692-2271 or (614) 692-3191.

7.8 Legacy. This commercial item description is a replacement for MIL-P-54001 and MS49006 for all federal agencies (MIL-P-54001 is canceled as of 9 March 1973 and MS49006 is canceled as of 15 November 1972 and copies of these documents are available online at http://assist.quicksearch.dla.mil or from the U.S. Government Printing Office, 732 North Capital Street, NW, Washington D.C. 20401.)

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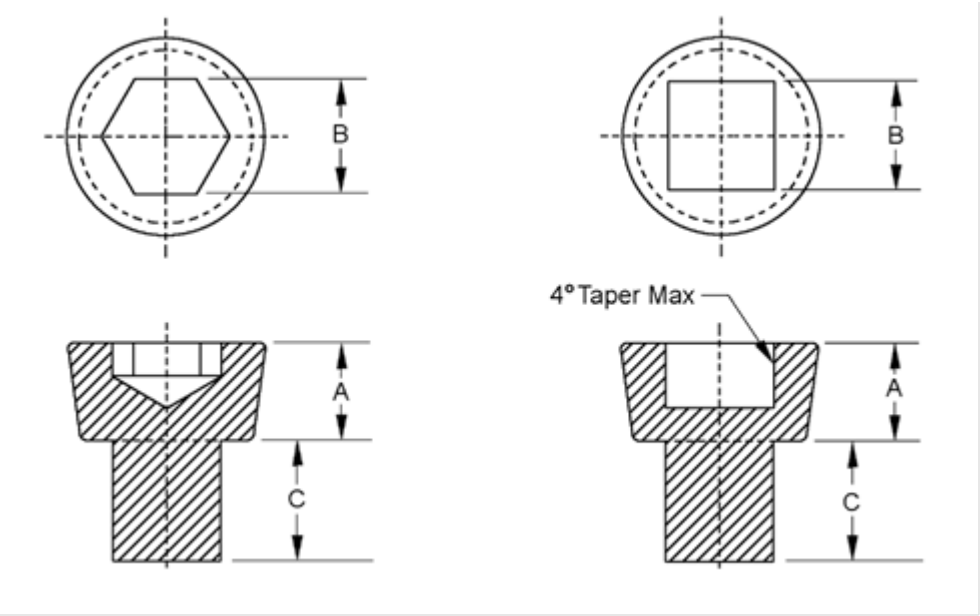
TABLE II. Pipe plugs, magnetic - headless. [5/]

Dash numbers [1/,2/,3/]		Nominal size – NPTF	A		B		C	Lift [6/]	Head style (recessed)
Type I	Type II		Min.	Max.	Min.	Max.	Max.	Min.	
01	02	0.125 - 27	0.290	0.310	0.188	0.193	0.375	4 oz	Hex socket
03	04	0.250 - 18	0.445	0.475	0.250	0.255	0.375	4 oz	Hex socket
05	06	0.375 - 18	0.445	0.475	0.313	0.318	0.375	4 oz	Hex socket
07	08	0.500 - 14	0.590	0.630	0.375	0.390	0.375	4 oz	Hex socket
09	10	0.750 - 14	0.600	0.640	0.563	0.568	0.500	6 oz	Hex socket
11	12	1.000 - 11 1/2	0.750	0.790	0.625	0.630	0.500	6 oz	Hex socket
13	14	1.250 - 11 1/2	0.800	- - - -	0.759	0.780	0.500	8 oz	Square socket
	16	1.500 - 11 1/2	0.830	- - - -	0.759	0.780	0.500	8 oz	Square socket
	18	2.000 - 11 1/2	0.890	- - - -	0.884	0.900	0.625	10 oz	Square socket
	20	2.500 - 8	1.070	- - - -	1.137	1.160	0.625	10 oz	Square socket
	22	3.000 - 8	1.130	- - - -	1.391	1.410	0.625	10 oz	Square socket
	24	4.000 - 8 NPT	1.220	- - - -	2.022	2.040	0.625	10 oz	Square socket

NOTES:

1. Types, see 3.0.
2. Materials, see 3.2.1.
3. Finish, see 3.2.3.
4. Threads, see 3.3
5. Dimensions and tolerances. All dimensions are in inches. Figure I and table II are for identification purposes only and are not intended to restrict designs or shapes of plugs not dimensioned herein. Dimensions and tolerances not shown shall be in accordance with SAE J531 for sizes up to and including one (1) inch and in accordance with ASME B16.14 for sizes greater than one (1) inch.
6. Magnetic lift rating, see 3.4.

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NOTE: For dimensions, see table I.

FIGURE 1. Pipe plugs, magnetic - headless.

7.9 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

MILITARY INTERESTS

Custodians:
Army - AR
Navy - YD
Air Force - 99
DLA - CC

Review activities:
Army-AT
Navy – SA,MC
Air Force - 71

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing activity:
DLA - CC

Project 4730-2014-001

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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